SOC Lab: Failed Login Detection using Microsoft Sentinel

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Course / Platform: Microsoft Sentinel Lab / Azure

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1. Project Overview

This project demonstrates a mini SOC lab setup where failed login attempts on a Windows VM are collected, monitored, and detected using **Microsoft Sentinel**.

The goal is to detect **brute-force login attempts** using KQL queries and generate incidents in Sentinel.

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2. Tools & Technologies

- **Microsoft Azure** – Cloud environment
- **Microsoft Sentinel** – SIEM platform
- **Windows VM** – Source of logs
- **KQL (Kusto Query Language)** – For detection queries

3. Step-by-Step Lab Execution

Step 1: Create a Windows VM in Azure

- Log in to Azure portal
- Go to **Virtual Machines → Add**
- Fill details:
    - Name: SOC-Lab-VM
    - Resource Group: soc-lab-rg
    - OS: Windows Server 2019
- Start the VM and connect via RDP

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Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure subscription 1

Resource group * : (New) soc-lab-rg
Create new

Instance details

Virtual machine name * : soc-lab-vm

Region * : (Asia Pacific) Central India
Deploy to an Azure Extended Zone

Availability options : No infrastructure redundancy required

Security type : Standard

Image * : Windows Server 2022 Datacenter - x64 Gen2 (free services eligible)
See all images | Configure VM generation

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Image * : Windows Server 2022 Datacenter - x64 Gen2 (free services eligible)
See all images | Configure VM generation

This image is compatible with additional security features. Click here to swap to the Trusted launch security type.

VM architecture : Arm64 / x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount

You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. Learn more

Size * : Standard_B2as_v2 - 2 vcpus, 8 GiB memory ($42.63)
See all sizes

Enable Hibernation

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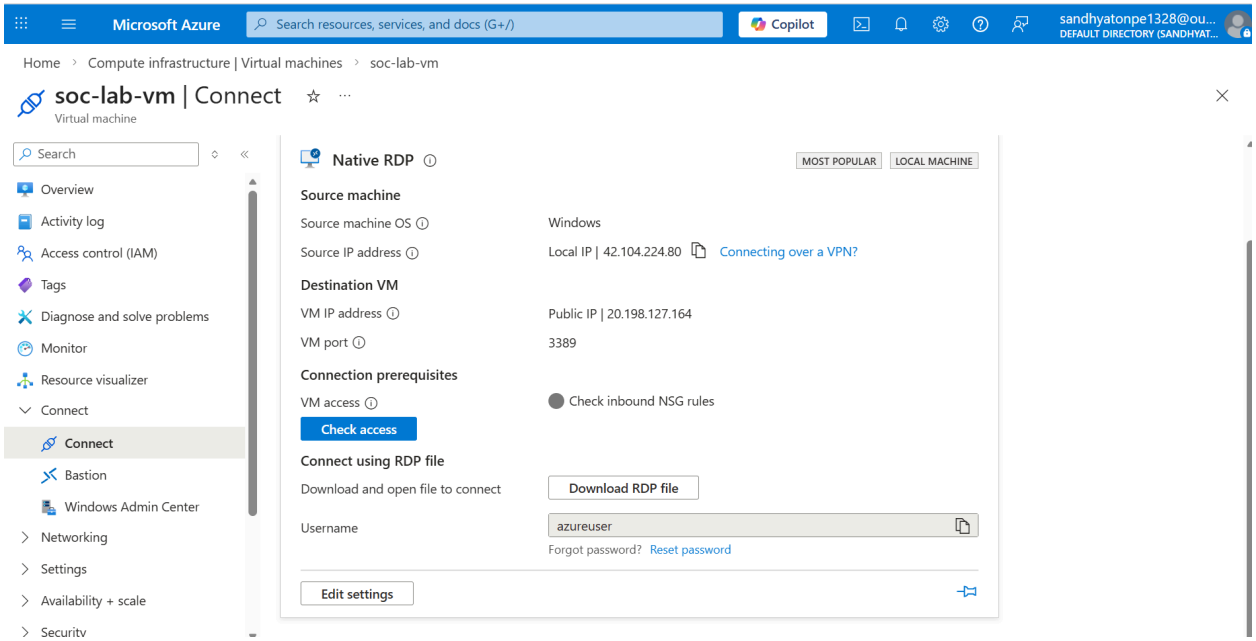
Create | Reservations | Manage view | Refresh | Export to CSV | Open query | ... | Group by none

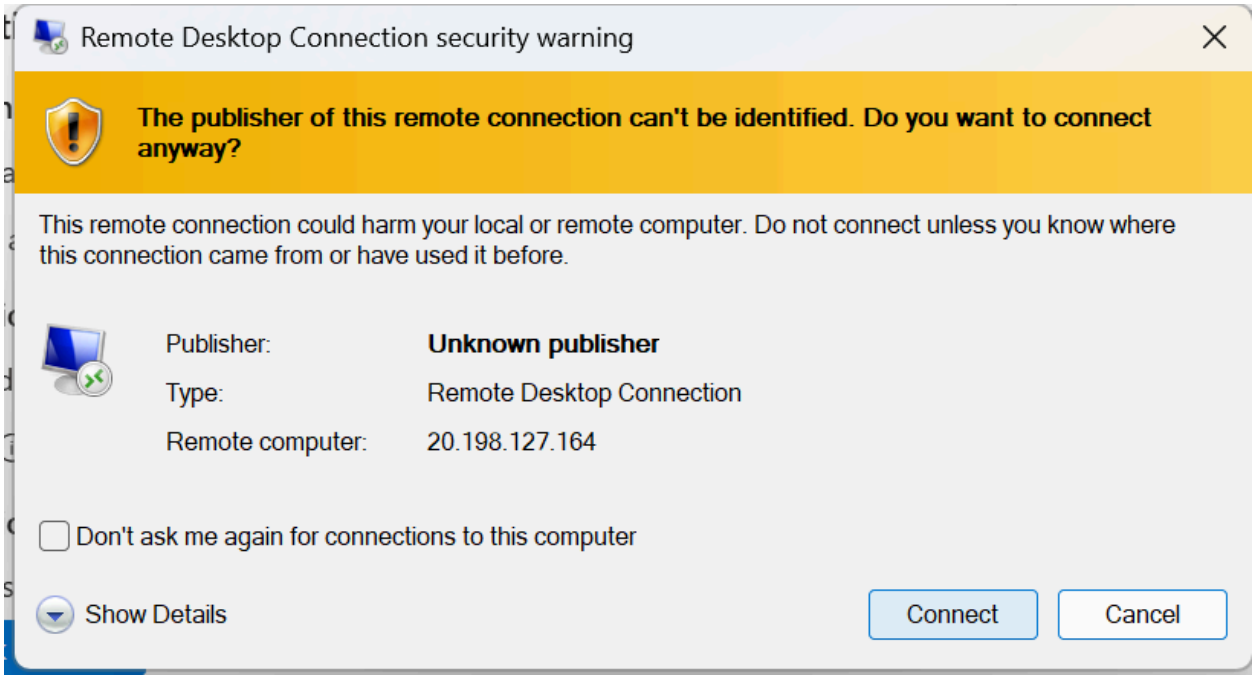
Filter for any field... | Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size
soc-lab-vm	Azure subscript...	soc-lab-rg	Central India	Running	Windows	Standard_B2as_...

Showing 1 - 1 of 1. Display count: auto

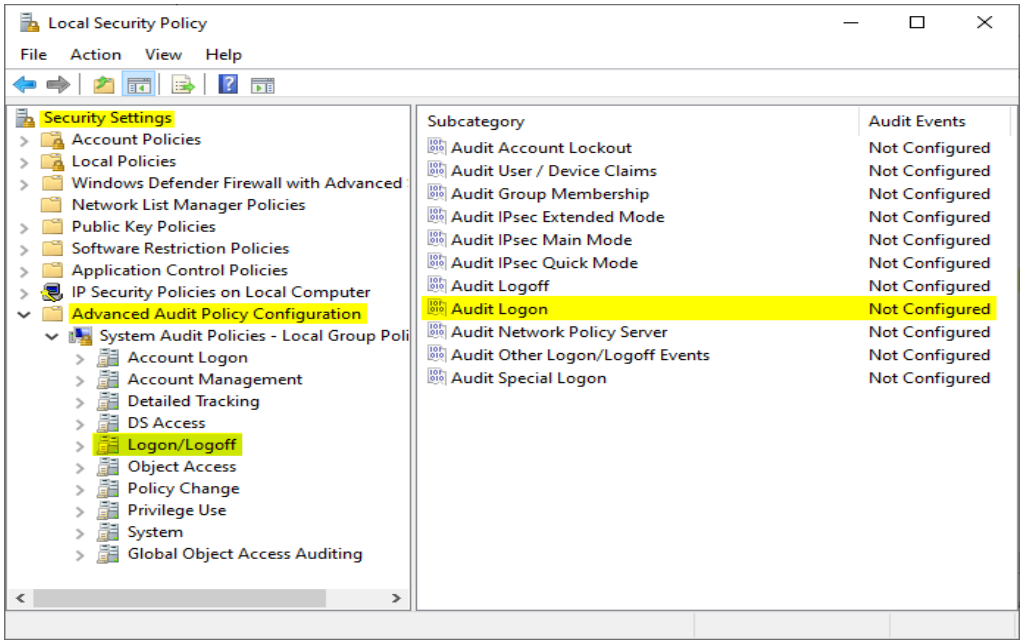
Give feedback

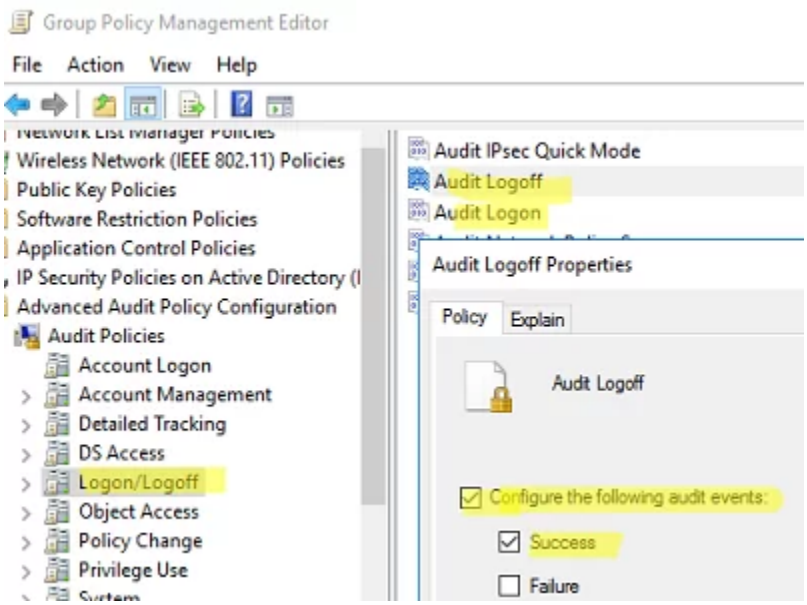




Step 2: Enable Audit Policy on VM

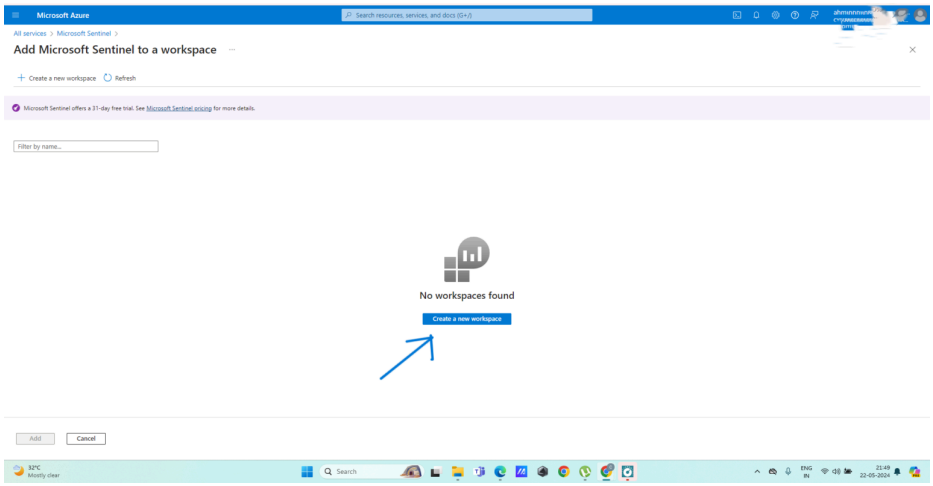
- Open **Local Security Policy → Advanced Audit Policy Configuration → Logon/Logoff → Audit Logon → Failure**
- This ensures failed login attempts are logged

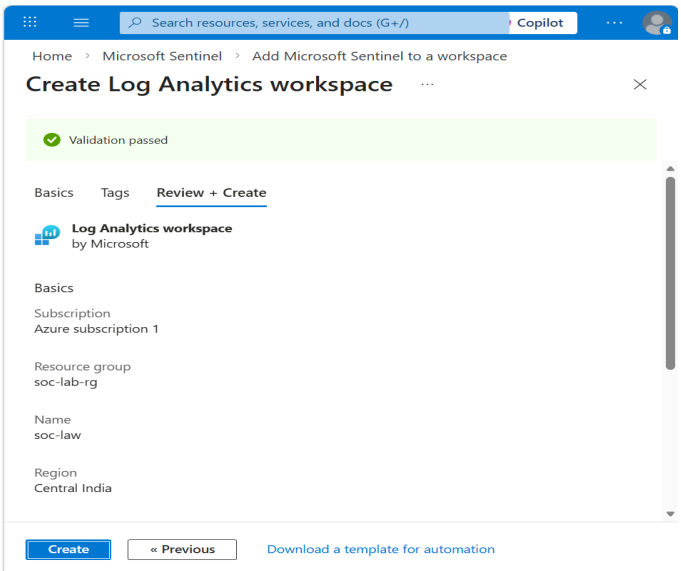




Step 3: Create a Microsoft Sentinel Workspace

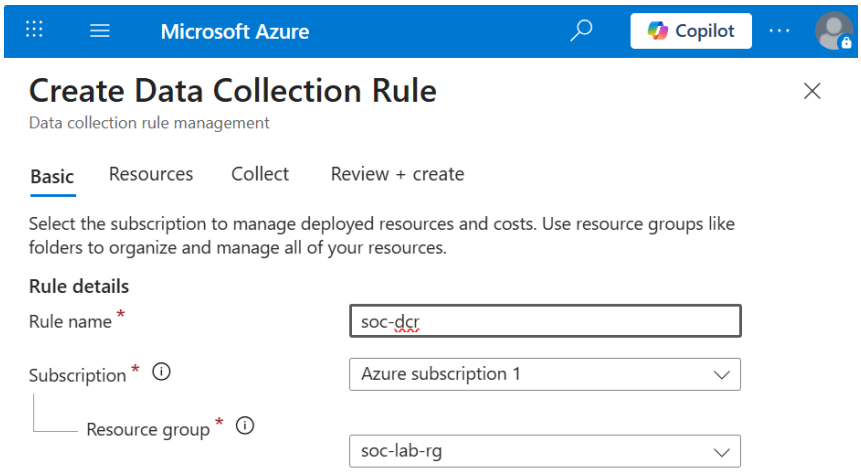
- Navigate to **Microsoft Sentinel → Add**
- Create a new workspace (or attach to existing)
- Name workspace: SOC-Lab-Sentinel
- Enable **Log Analytics**

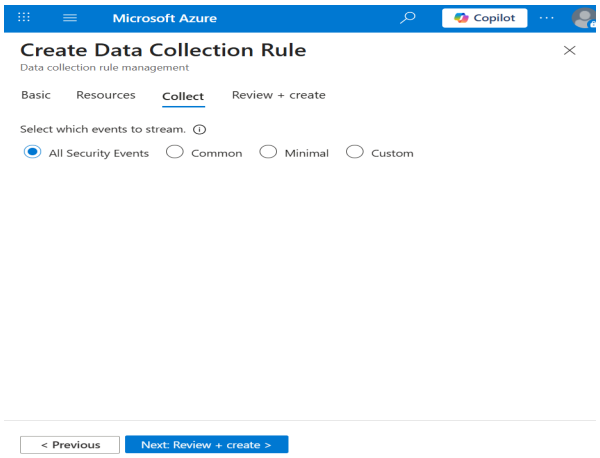


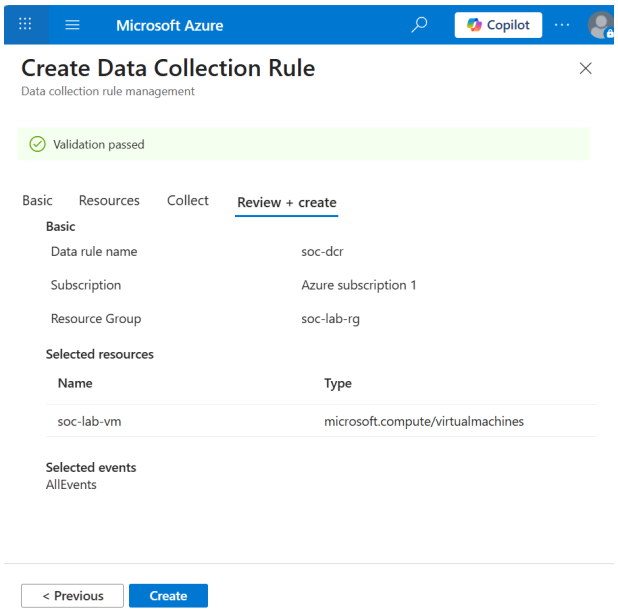


Step 4: Connect Windows VM Logs to Sentinel

- Go to **Data connectors → Windows Security Events → Connect**
- Select **VM SOC-Lab-VM**
- Apply settings → Logs start flowing (wait 5–10 min)



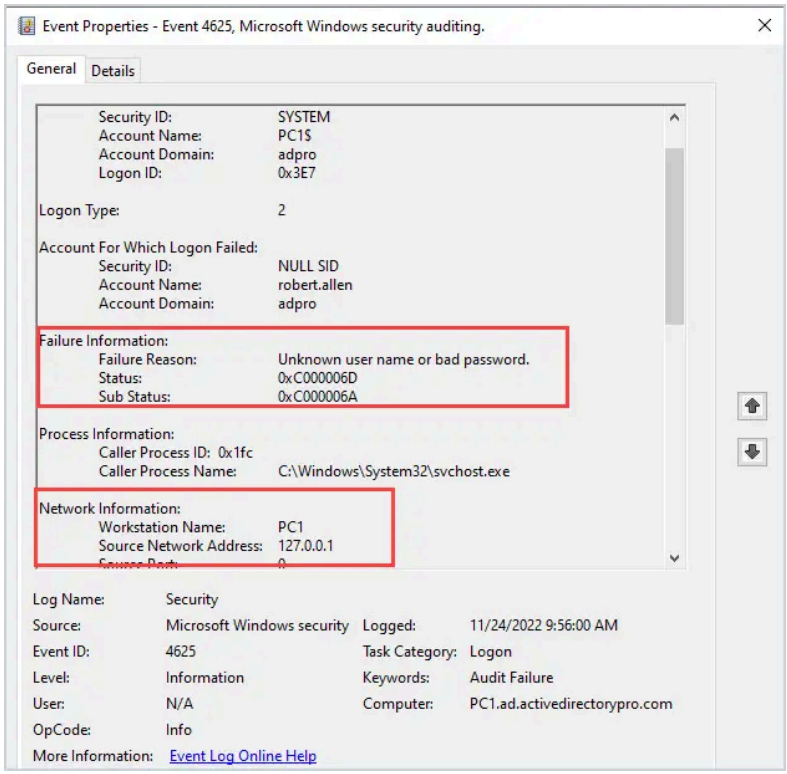




Step 5: Generate Failed Login Events

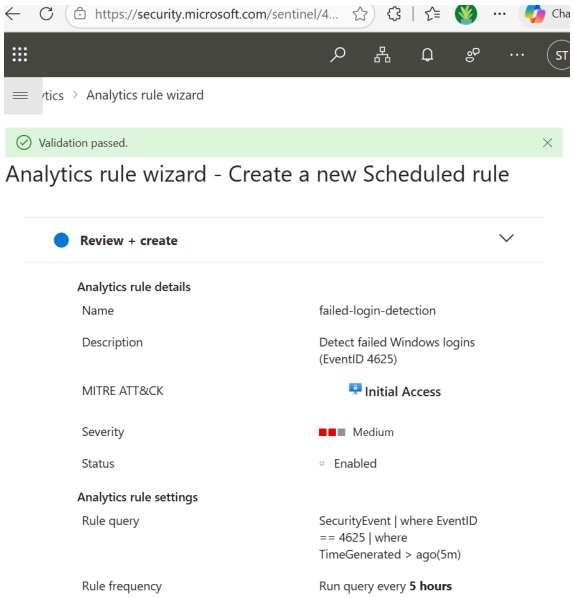
- On the Windows VM, try wrong login attempts

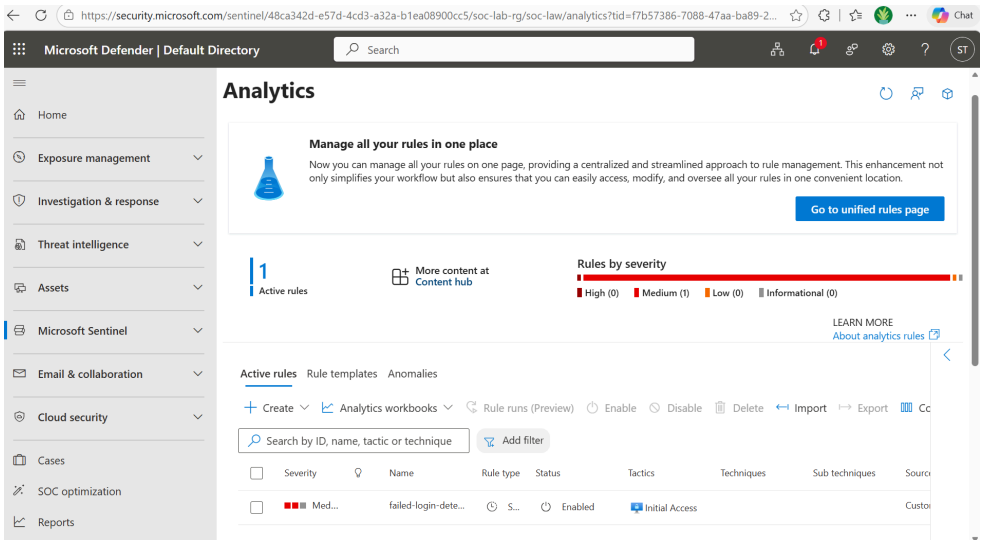
- Ensure EventID 4625 appears in **Event Viewer → Security logs**



Step 6: Create Custom Analytics Rule in Sentinel

- Go to **Analytics → + Create → Scheduled query rule**
- Fill details:
  - Rule name: soc-dcr
  - Subscription: Azure subscription 1
  - Resource group: soc-lab-rg
- Next → **Collect** → Select **All Security Events**
- Next → **Resources** → Add **SOC-Lab-VM** → Apply
- Next → **Review + Create → Create**
- Wait 5–10 minutes for logs and incidents

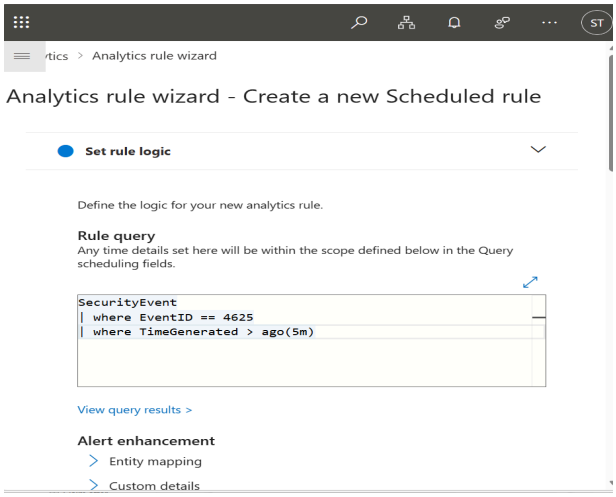




Step 7: KQL Query to Detect Failed Logins

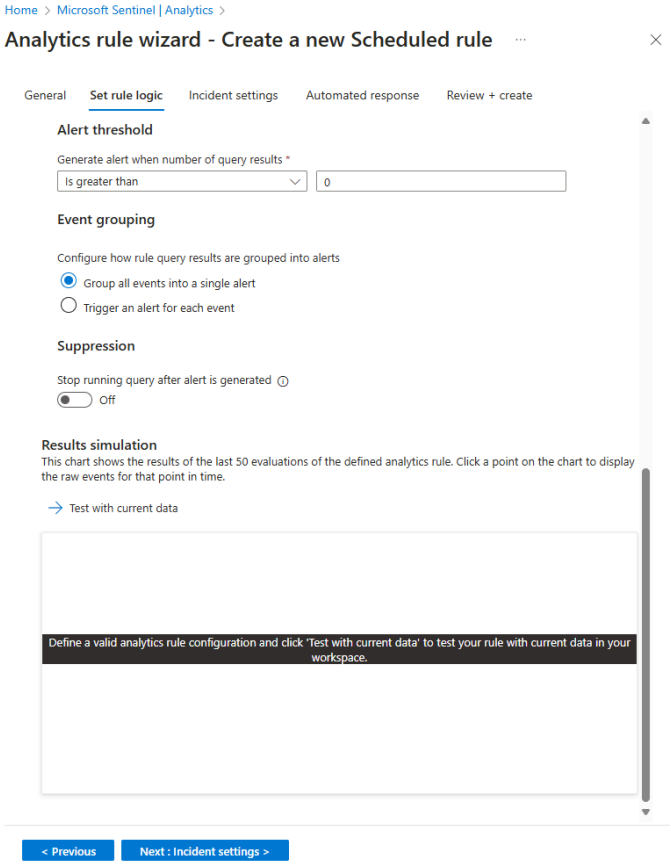
```
SecurityEvent
| where EventID == 4625
| where TimeGenerated > ago(1h)
```

- This query detects all failed login attempts in the last hour



Step 8: Test Analytics Rule

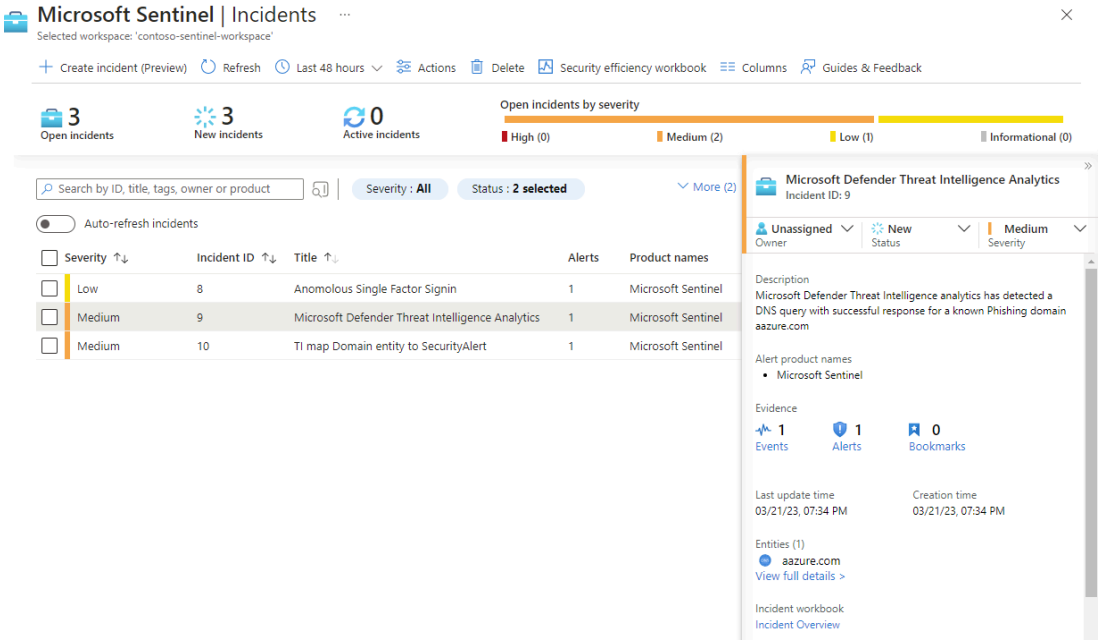
- If incidents don't appear automatically:
  - Go to **Analytics → select rule → Run query now**
  - Ensure incidents are generated based on logs

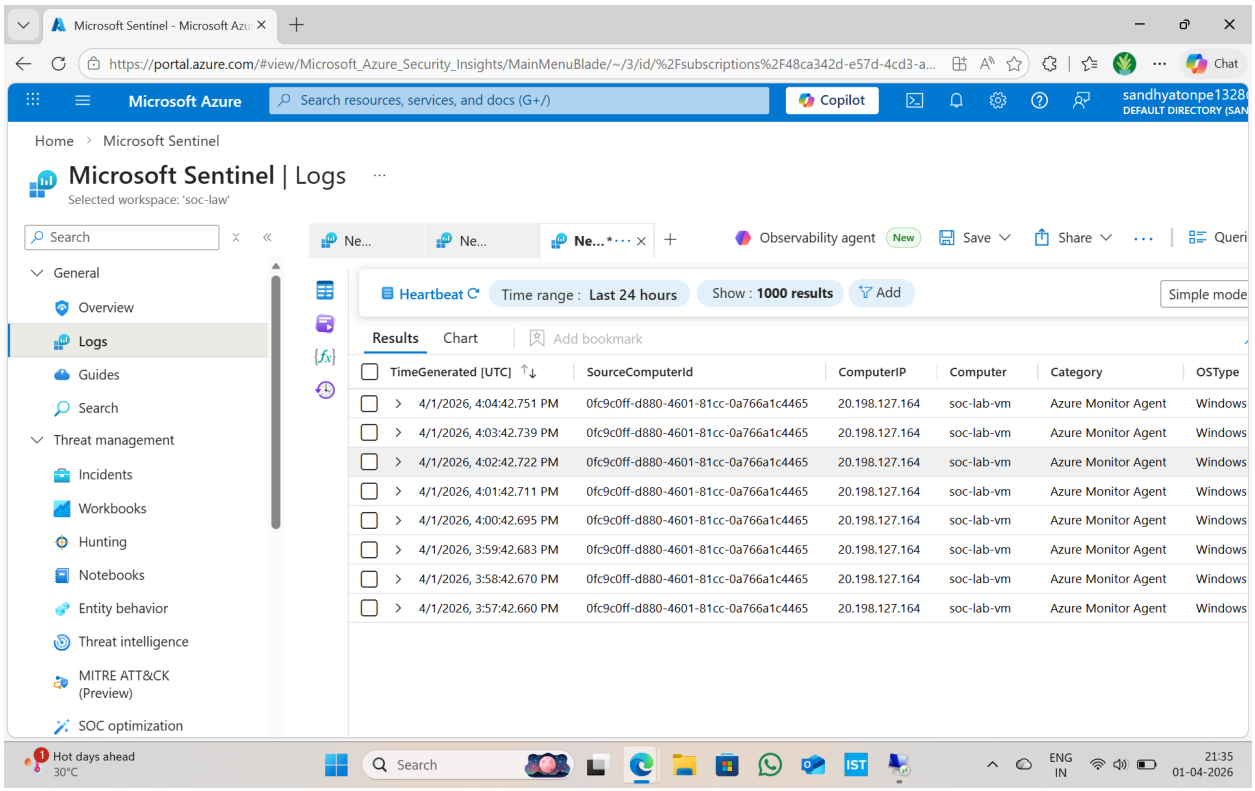


Step 9: Verify Incidents

- Go to **Incidents** in Sentinel

- Validate failed login attempts trigger the analytics rule and appear as incidents





4. Architecture Diagram

- VM → generates failed login events
- Sentinel → collects logs via Data Connector
- Analytics Rule → runs KQL query to detect EventID 4625
- Incidents → generated in Sentinel

Flow:

VM → Logs → Sentinel → Analytics Rule → Incident